

NTIRE 2026 Nighttime Image Dehazing Challenge

Technical Report — Team Innovaide

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March 2026

Summary. We fine-tune **FFA-Net** (Feature Fusion Attention Network) from ITS daytime pretrained weights onto the NTIRE 2026 nighttime haze training data, and apply **8-fold Test-Time Augmentation (TTA)** at inference. Our final submission achieves **PSNR 20.452 dB**, **SSIM 0.7561**, and **LPIPS 0.2423** on the competition test set.

Team Details

Team name	Innovaide
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Code	https://github.com/ashvp/Dehazing-NTIRE-2026/tree/main

Results

PSNR (dB)	SSIM	LPIPS
20.452	0.7561	0.2423

PSNR and SSIM are higher-is-better; LPIPS (perceptual similarity via deep features) is lower-is-better.

Approach Overview

Nighttime haze differs significantly from daytime haze — scenes have low ambient light, artificial light sources, glow and halo artifacts around point lights, and non-uniform brightness. Training a dehazing model from scratch on the ≈ 25 provided pairs is insufficient.

Our strategy was straightforward:

1. Start from FFA-Net pretrained on indoor daytime haze (ITS), so the model already understands haze structure.
2. Fine-tune on the NTIRE 2026 nighttime training data with heavy augmentation.
3. At inference, run the model on 8 geometrically augmented versions of each input and average the results (TTA).

Architecture: FFA-Net

FFA-Net (Feature Fusion Attention Network) [1] is a CNN-based single image dehazing model. It passes the input through three sequential *feature groups*, each consisting of 19 residual blocks with dual attention. The outputs of all three groups are fused via a learned attention weighting, and a global residual skip adds back the original hazy input so the model only needs to predict the correction.

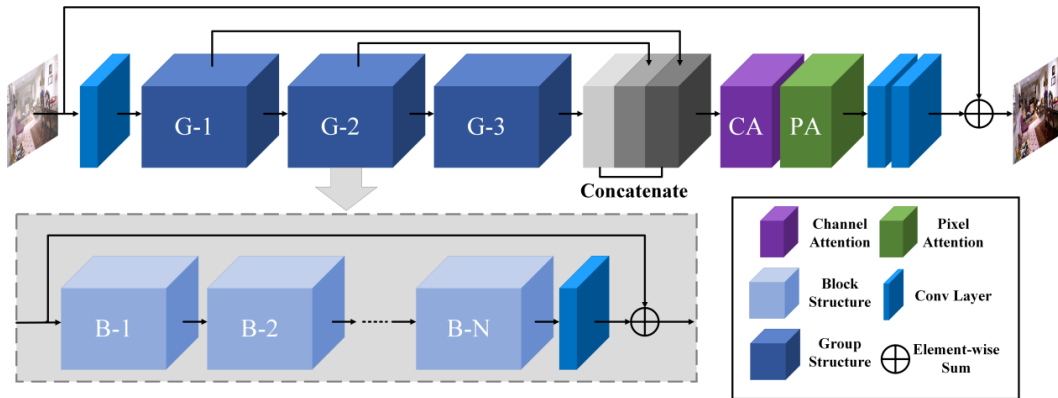


Figure 1: FFA-Net pipeline. Three feature groups process the image in sequence; their outputs are concatenated and refined by Channel Attention (CA) and Pixel Attention (PA) before reconstruction.

Channel Attention and Pixel Attention

Within each residual block, FFA-Net applies two types of attention in sequence:

Channel Attention (CA) asks: which feature channels are most important? It computes a global average over the spatial dimensions and passes it through a small fully-connected network to produce per-channel scaling weights. This lets the model selectively emphasise features related to haze density.

Pixel Attention (PA) asks: which spatial locations matter most? It applies a 1×1 convolution to produce a per-pixel weight map, allowing the network to focus on heavily hazed regions while leaving clearer areas mostly unchanged.

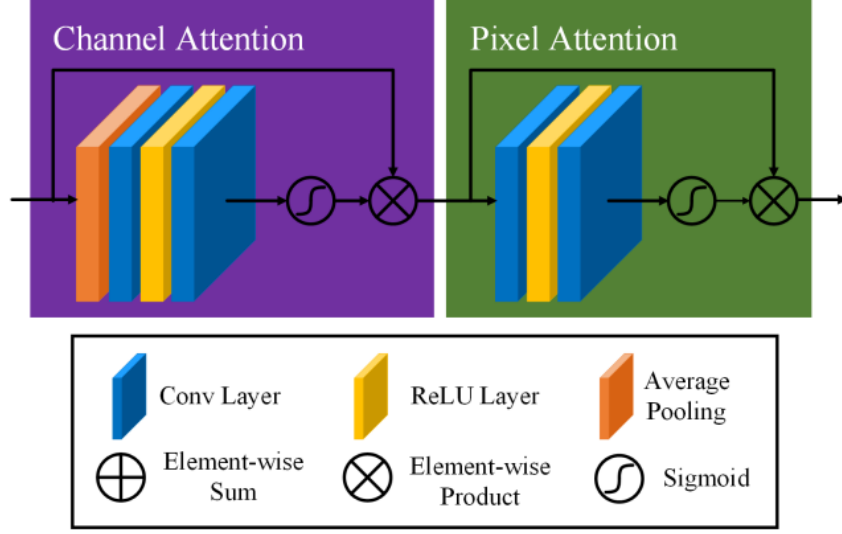


Figure 2: figure

Inside each residual block: Channel Attention rescales channels globally; Pixel Attention rescales spatially, pixel by pixel.

Attention-Based Feature Fusion

After the three groups produce their feature maps, a second channel attention module receives their concatenation and learns per-group weights. The weighted sum passes through a final pixel attention and 3×3 convolution to reconstruct the clean image.

Training Details

Data Augmentation

With only ≈ 25 training pairs, preventing overfitting required aggressive augmentation. Per epoch, 50 random 256×256 crops were extracted per image (effective 1,250 samples/epoch), with random horizontal flip, vertical flip, and $90/180/270^\circ$ rotations applied to each crop.

Loss Function

We used the **Charbonnier loss**, a smooth approximation of L_1 :

$$\mathcal{L} = \frac{1}{N} \sum_i \sqrt{(\hat{y}_i - y_i)^2 + \varepsilon}, \quad \varepsilon = 10^{-6} \quad (1)$$

Compared to MSE it is less sensitive to outlier pixels and produces sharper results; compared to L_1 it is differentiable everywhere, giving smoother gradients.

Optimiser and Scheduler

Adam optimiser ($\beta_1 = 0.9$, $\beta_2 = 0.999$), initial learning rate 5×10^{-5} with a 3-epoch linear warmup followed by cosine annealing to near-zero. The lower learning rate (compared to training from scratch) was chosen to avoid destabilising the pretrained weights during early fine-tuning. Gradient clipping was applied with max norm 1.0.

Pretrained Initialisation

The model was initialised from FFA-Net weights pretrained on the RESIDE Indoor Training Set (ITS) — synthetic indoor daytime haze. Even across the day-to-night domain gap, these weights provided strong haze-removal priors that significantly outperformed training from scratch, as confirmed by ablation.

Test-Time Augmentation (TTA)

At inference, we apply 8-fold TTA. For each input, the model is run on 8 geometric variants of the image; each prediction is inverse-transformed back to the original orientation, and the 8 results are averaged pixel-by-pixel. This smooths any directional biases from training.

#	Augmentation	Inverse
1	Identity (original)	None
2	Horizontal flip	Horizontal flip
3	Vertical flip	Vertical flip
4	Rotate 90°	Rotate 270°
5	Rotate 180°	Rotate 180°
6	Rotate 270°	Rotate 90°
7	H-flip + Rotate 90°	Rotate 270° + H-flip
8	V-flip + Rotate 90°	Rotate 270° + V-flip

Other Architectures Evaluated

Restormer

Restormer [2] is a Transformer-based restoration model. Its key module, **Multi-Dconv Head Transposed Attention (MDTA)**, computes self-attention across channels rather than spatial positions, making it efficient at high resolution. A **Gated Dconv Feed-Forward Network (GDFN)** adds spatial context via gated depth-wise convolutions. The full model is a U-Net across four scales with skip connections.

We fine-tuned Restormer and used it as a teacher in distillation experiments. Its larger parameter count ($\approx 26\text{M}$) made it impractical as the final submission given the small dataset.

NAFNet

NAFNet [3] (Nonlinear Activation Free Network) removes all nonlinear activations and replaces them with a **SimpleGate** — the channel dimension is split in two and the halves are multiplied element-wise. This acts as a learned gating mechanism without ReLU or GELU, making the model very fast and stable. It also uses **Simplified Channel Attention (SCA)** — just global average pooling followed by a single convolution — and Group Normalization for stability at small batch sizes.

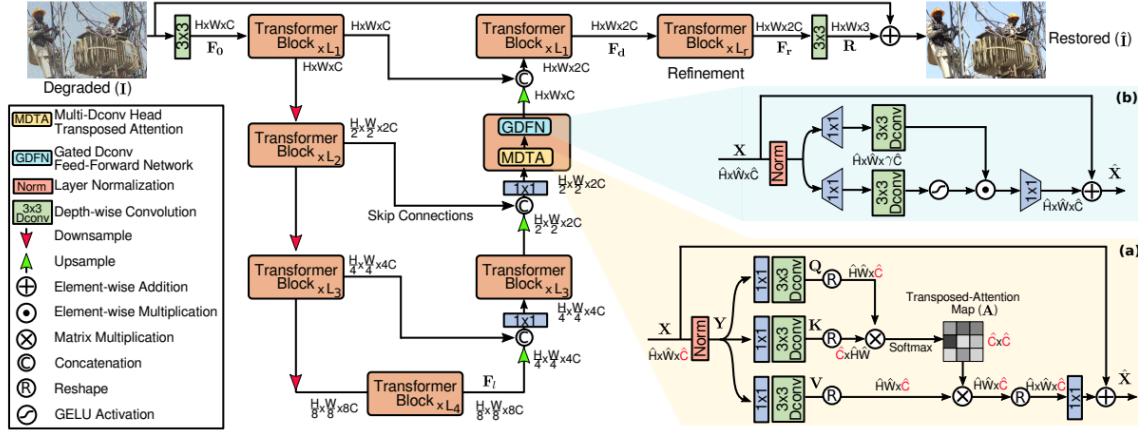


Figure 3: figure

Restormer U-Net. MDTA computes transposed attention across channels; GDFN gates the feed-forward path with depth-wise convolutions.

Despite its speed, NAFNet failed to converge on the nighttime haze domain. We attribute this to its simplified gating being insufficient for the complex combination of haze, low-light, and artificial light artifacts, combined with the very small dataset.

All Experiments Tried

Approach	PSNR (dB)	Notes
FFA-Net from scratch	—	No pretrained benefit
Restormer	19.06	Outputs had box artifacts
Restormer + TTA	19.67	Reduced box artifacts
Restormer + FFA-Net ensemble	20.04	Averaged outputs blurred detail
Restormer + FFA-Net ensemble + TTA	20.24	Averaged outputs
NAFNet	—	Failed to converge
Restormer + FFA-Net + TTA + GAN	20.41	GAN artifacts hurt pixel fidelity
NightFFA++ (custom)	—	Too small a dataset
FFA-Net fine-tuned + TTA	20.452	Final submission

NightFFA++: Custom Architecture

NightFFA++ was a custom model designed specifically for nighttime dehazing. Despite its richer design, it did not outperform the fine-tuned FFA-Net baseline within our training budget.

Design

Multi-scale U-Net with FFA blocks. Three encoder/decoder scales connected by strided and transposed convolutions, with skip connections at each scale. An **illumination estimation head** at the bottleneck produces a single-channel brightness map that is upsampled and

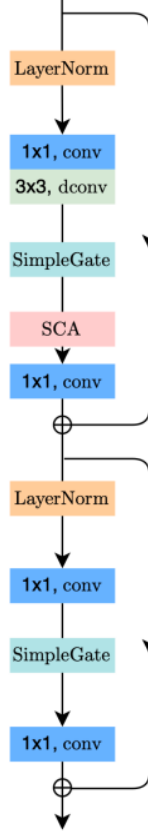


Figure 4: NAFNet block. SimpleGate ($x_1 \cdot x_2$ after channel split) replaces all nonlinear activations; SCA provides lightweight channel attention.

concatenated at each decoder stage — explicitly conditioning the dehazing on the estimated scene brightness, which is critical for nighttime images.

RGB + HSV early fusion. The hazy input is simultaneously converted to HSV colour space and both representations (6 channels total) are fused in the first projection layer. The Saturation (S) and Value (V) channels directly encode haze-induced saturation loss and brightness suppression, potentially helping the network disentangle illumination from haze density.

Composite loss function:

$$\mathcal{L} = \mathcal{L}_{\text{Charb}} + 0.2 \mathcal{L}_{\text{SSIM}} + 0.05 \mathcal{L}_{\text{perc}} + 0.05 \mathcal{L}_{\text{freq}} + 0.2 \mathcal{L}_{\text{exp}} + 0.15 \mathcal{L}_{\text{distill}} \quad (2)$$

where $\mathcal{L}_{\text{distill}}$ is an L_1 loss against a frozen fine-tuned Restormer teacher.

Why It Did Not Work

The ≈ 25 -pair dataset was too small to train a larger model without overfitting. The composite loss required more hyperparameter tuning than was feasible in the competition timeframe. The Restormer teacher itself had limited nighttime performance, so the distillation signal introduced noise rather than useful guidance.

Technical Summary

Property	Value
Architecture	FFA-Net (gps = 3, blocks = 19), $\approx 4.68\text{M}$ parameters
Pretrained init	ITS weights (<code>its_train_ffa_3_19.pk</code>)
Training time	≈ 2.5 hours
Epochs	30 (best checkpoint by validation PSNR)
Loss	Charbonnier ($\varepsilon = 10^{-6}$)
Optimiser	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$), LR = 5×10^{-5}
LR schedule	Cosine annealing with 3-epoch linear warmup
Batch / patch size	4 / 256×256 , 50 random crops per image
Mixed precision	Yes (AMP)
Gradient clipping	Max norm 1.0
Inference	8-augmentation TTA (≈ 1 s per image)
GPU	2 $\times$ NVIDIA T4 (DataParallel, Kaggle)
Extra data	None — competition data only
Diffusion / GAN	No
Quantization	No

Comments and Future Directions

The NTIRE 2026 Nighttime Dehazing Challenge is a well-motivated benchmark. Nighttime haze is harder than daytime dehazing due to non-uniform illumination, light source interference, and reduced signal-to-noise in dark regions.

Suggestions for future editions:

- Provide a larger paired training set — 25 pairs is a significant constraint on model complexity.
- Consider a perceptual quality sub-track (LPIPS-ranked) alongside PSNR/SSIM.
- An efficiency-constrained track for real-time edge deployment would be valuable.
- Blind evaluation on fully held-out real-world nighttime data would better test generalization.

References

- [1] X. Qin, Z. Wang, Y. Bai, X. Xie, and H. Jia, “FFA-Net: Feature Fusion Attention Network for Single Image Dehazing,” *AAAI*, 2020.
- [2] S. W. Zamir *et al.*, “Restormer: Efficient Transformer for High-Resolution Image Restoration,” *CVPR*, 2022.
- [3] L. Chen *et al.*, “Simple Baselines for Image Restoration (NAFNet),” *ECCV*, 2022.